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Education

University of California, Los Angeles (UCLA) — Expected June 2028
Bachelor of Science in Computer Science — *Los Angeles, CA*

Experience

Software Development Engineer Intern (Incoming) — Sep 2026 – Dec 2026
Amazon (AWS) — *Boston, MA*

Software Engineering Intern — June 2026 – Present
Google (YouTube) — *San Bruno, CA*

- Engineered 10-bit software AV1 HDR video playback for the YouTube Android app via low-level ARM/NEON SIMD vectorization in C++ to improve the video quality and playback performance for millions of users.
- Reduced CPU utilization by 50% on 1080p HDR playback under dav1d software decoding, establishing dynamic resolution thresholds that adapt to device thermal and battery state to conserve power on Android devices.
- Built a Java-based shader effect debug UI and OpenGL video pipeline to accelerate developer iteration speed.
- Integrated custom fragment shaders to allow video effects across thousands of DRM-protected shows and movies.
- Resolved memory leaks, migrated deprecated code, and improved format filtering across ExoPlayer and YouTube.

Open Source Contributor — Jan 2026 – Present
XXXXXXXX — *Remote*

- Built a shared dynamics core powering limiter and expander audio filters to cut duplicate code by 20%.
- Integrated ggml and OpenCV dependencies into VLC's build system to enable on-device model inference.
- Ported 3 legacy OpenCV modules from C to C++ and upgraded the SAM2 filter to SAM3 to allow text prompts.
- Cut the final VLC binary size by 5% using linker-level dead-code stripping to reduce unnecessary dependencies.

Software Engineering Intern — June 2025 – Sep 2025
XXXXXXXX — *Remote*

- Integrated a SAM2 model module with OpenCV to enable real-time segmentation of 100+ concurrent objects.
- Implemented YuNet-based ML face detection to allow tracking for 100+ concurrent faces in surveillance systems.
- Built fast video-to-GIF conversion with five dithering algorithms for customizable quality and performance.

Projects

XXXXXXXXXXXX | *C++, Assembly/SIMD, FFmpeg, Coding Theory, Compression* — Feb 2026 – Present

- Popular C++ tool (890+ stars, 110+ forks) that encodes local files into YouTube videos for storage.
- Achieved a 100x speed-up using inline Assembly, SIMD intrinsics, and OpenMP parallelization.
- Designed lightweight Qt 6 UI and libsodium encryption, enhancing user experience and security.
- Uses robust CRC/Wirehair error-correction to bypass lossy video compression algorithms.
- Reached #5 on Hacker News (Y Combinator) within 12 hours, generating thousands of views.
- Includes technical explanation video with over 3.3M impressions, educating over 270k viewers.

XXXX | *Java, Spring Boot, TypeScript, CI/CD* — June 2020 – Present

- Popular Java framework (130+ stars, 10 forks) for building fast and portable media applications.
- Integrates with VLC/mpv with a documented image-processing pipeline API for easier adoption.
- Uses Spring Boot to stream sub-100ms latency audio to usable TypeScript/Howler.js frontend.
- Configured CI/CD pipeline with TeamCity using Oracle Cloud to automate software releases.
- Engineered native media player via JavaCV that is 5x faster than most pure-Java players.

Technical Skills

Languages: Java, C/C++, OCaml, JavaScript/TypeScript, Assembly
Frameworks & Libraries: Spring, Hibernate, FFmpeg, OpenCV, libVLC, React, Next.js, Tailwind
Technologies: Git, MongoDB, AWS, Google Cloud, Oracle Cloud, CMake, Meson, OpenGL, ExoPlayer, Android
Awards: Dean's Honor List